THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 10: **Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): The image formed by a plane mirror is virtual. Reason (R): It cannot be obtained on a screen. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. The angle between a ray and a plane mirror is 35 degrees. If the mirror is rotated by 15 degrees so that the angle of incidence increases, the new angle between the incident and the reflected ray is: 1

(a) 110 degrees **(b)** 100 degrees **(c)** 140 degrees **(d)** 70 degrees

3. Light incident on a smooth metal surface at 90 degrees to the surface is reflected at: 1

(a) 0 degrees
(b) 90 degrees to the surface (back along normal)
(c) 45 degrees
(d) 180 degrees

4. Assertion (A): A concave mirror is used as a shaving mirror. Reason (R): It forms an enlarged erect image of the face when held close. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Two plane mirrors are at right angles. A ray hits the first at an angle of incidence of 25 degrees. The angle of incidence on the second is: 1

(a) 90 degrees **(b)** 65 degrees **(c)** 50 degrees **(d)** 25 degrees

6. An object is placed at the focus of a concave mirror. The reflected rays are: 1

(a) parallel **(b)** divergent **(c)** convergent **(d)** none

7. Assertion (A): Convex mirrors are used as rear-view mirrors in vehicles. Reason (R): They always form real and enlarged images. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A man stands 3 m in front of a plane mirror and a boy stands 2 m behind the man. The distance of the boy's image from the man is: 1

(a) 5 m **(b)** 3 m **(c)** 8 m **(d)** 2 m

9. A student sees in a concave mirror a diminished inverted real image. The object is: 1

(a) at C **(b)** at F **(c)** beyond C **(d)** between F and P

10. Assertion (A): A convex lens can start a fire using sunlight. Reason (R): It converges parallel rays to a point. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A ray incident on a mirror at 30 degrees to the surface is reflected. If the mirror turns through 10 degrees, find the possible new angle of incidence. 2

12. Why does a plane mirror show an image of the same size even though the image looks 'far away'? 2

13. Why is a convex lens also called a converging lens, and where does a beam parallel to its axis meet? 2

14. A person walks away from a plane mirror at 3 m/s. Find the speed at which the image moves relative to the person. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two plane mirrors are inclined at an angle theta. If 7 images are formed, find theta. 3

16. A student uses a concave mirror as a make-up mirror and sees an enlarged face. When she moves away, the face appears inverted. Explain. 3

17. Compare real image formation in a camera and an eye in terms of lens and screen. 3

OR

An object is 15 cm in front of a plane mirror. Find the distance of the image from the mirror and from the object. If the object moves 4 cm towards the mirror, find the new distance between object and image. 3

18. Explain the use of a reflector behind a bulb in a torch with a diagram description. 3

OR

If a mirror is curved outwards, how do you predict the nature of the image and its size? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the formation of the image of an object placed in front of a concave mirror at six positions. For the object at 30 cm from a mirror of focal length 10 cm state the position, nature and size. 5

OR

Describe in detail the working of the eye as an optical instrument and explain defects of vision, with remedies and lens power ideas. 5

20. Two plane mirrors are placed at right angles. A ray strikes the first mirror at an angle of incidence of 35 degrees. Find the angle of incidence at the second mirror and show that the emergent ray is parallel to the incident ray. 5

OR

Discuss how reflection by smooth and rough surfaces is useful in everyday life including the visibility of objects, road signs, projection screens and mirrors. 5

END OF QUESTION PAPER — Total: 40 Marks